

Complex Numbers

Modulus. $|z| = \sqrt{x^2 + y^2}$; $|z_1 z_2| = |z_1| |z_2|$; $|z_1/z_2| = |z_1|/|z_2|$;
 $|z^m| = |z|^m$; $z\bar{z} = |z|^2$; $z^{-1} = \bar{z}/|z|^2$

Conjugation. $\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2$; $\overline{z_1 z_2} = \bar{z}_1 \bar{z}_2$; $\overline{(z_1/z_2)} = \bar{z}_1/\bar{z}_2$;
 $\Re z = (z + \bar{z})/2$; $\Im z = (z - \bar{z})/(2i)$; $|\Re z|, |\Im z| \leq |z|$;
 $|z + w|^2 = |z|^2 + |w|^2 + 2\Re(z\bar{w})$

Triangle inequalities. $|z_1 + z_2| \leq |z_1| + |z_2|$;
 $|z_1 - z_2| \geq ||z_1| - |z_2||$; $|z_1 + z_2| \geq ||z_1| - |z_2||$

Polar form. $z = r(\cos \theta + i \sin \theta) = r e^{i\theta}$;
 $z^n = r^n(\cos(n\theta) + i \sin(n\theta)) = r^n e^{in\theta}$ (de Moivre); Euler:
 $e^{i\theta} = \cos \theta + i \sin \theta$

Complex Functions

Exponential. $e^z := e^x(\cos y + i \sin y)$ for $z = x + iy$

Logarithm. General: $\log z = \ln |z| + i(\arg z + 2k\pi)$, $k \in \mathbb{Z}$.
Principal branch: $\text{Log } z = \ln |z| + i \text{Arg } z$, $\text{Arg } z \in (-\pi, \pi]$,
continuous on $\mathbb{C} \setminus (-\infty, 0]$.

Power function. $z^\alpha := e^{\alpha \log z}$; finitely many values iff $\alpha \in \mathbb{Q}$,
otherwise infinite.

Trig functions. $\cos z := \frac{e^{iz} + e^{-iz}}{2}$; $\sin z := \frac{e^{iz} - e^{-iz}}{2i}$;
 $\cos^2 z + \sin^2 z = 1$; 2π -periodic.

Hyperbolic. $\cosh z := \frac{e^z + e^{-z}}{2}$; $\sinh z := \frac{e^z - e^{-z}}{2}$

Square root. $\pm \sqrt{r}(\cos(\theta/2) + i \sin(\theta/2))$; two branches,
discontinuous on $(-\infty, 0]$.

n-th roots. Solving $z^n = w = r_w e^{i\theta w}$:

$$z = r_w^{1/n} \exp\left(i \frac{\theta_w + 2\pi k}{n}\right), \quad k = 0, \dots, n-1$$

Solutions lie on a circle of radius $r_w^{1/n}$. Roots of unity ($z^n = 1$,
 $r_w = 1$) lie on unit circle.

Analytic Functions

Complex differentiability.

$$f'(z_0) = \lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0} = \lim_{h \rightarrow 0} \frac{f(z_0 + h) - f(z_0)}{h}$$

f is **analytic (holomorphic)** on D if complex differentiable
everywhere on D . **Goursat:** complex differentiable on $D \implies f'$
continuous, hence analytic.

Cauchy-Riemann equations. $f(x + iy) = u(x, y) + iv(x, y)$
differentiable at $z_0 \implies$

$$u_x = v_y, \quad u_y = -v_x$$

$\implies f'(z) = u_x + iv_x = v_y - iu_y$. Converse: $u, v \in C^1$ and CR holds
on $D \implies f$ analytic on D . e.g. $\bar{z}$: $u_x = 1 \neq v_y = -1$, not analytic
anywhere.

Harmonic functions. $\Delta u = u_{xx} + u_{yy} = 0$. $f = u + iv$ analytic
 $\implies u, v$ harmonic. v is a **harmonic conjugate** of u , unique up to
a constant; exists when D simply connected. *Finding v :* integrate
 $v_y = u_x$ and $v_x = -u_y$.

Constancy. f analytic and any of $|f|$, $\Re f$, $\Im f$, $\arg f$, $\bar{f}$ also
analytic $\implies f$ constant.

Fractional linear transformations (FLT / Möbius).

$$f(z) = \frac{az + b}{cz + d}, \quad ad - bc \neq 0$$

3 degrees of freedom; uniquely determined by 3 point-value pairs.
Composition $\leftrightarrow$ matrix multiplication; inverse is also a FLT.

Cross-ratio (FLT-invariant):

$$\frac{(w - w_1)(w_2 - w_3)}{(w - w_3)(w_2 - w_1)} = \frac{(z - z_1)(z_2 - z_3)}{(z - z_3)(z_2 - z_1)}$$

$$(z_0, z_1, z_2) \mapsto (0, 1, \infty): \quad w = \frac{(z_1 - z_2)(z - z_0)}{(z_1 - z_0)(z - z_2)}$$

$$(0, 1, \infty) \mapsto (z_0, z_1, z_2): \quad z = \frac{z_2(z_1 - z_0)w - z_0(z_1 - z_2)}{(z_1 - z_0)w - (z_1 - z_2)}$$

Line Integrals

Path integral. $\gamma(t) = (x(t), y(t))$, $a \leq t \leq b$:

$$\int_{\gamma} P dx + Q dy = \int_a^b [P(x(t), y(t)) \cdot x'(t) + Q(x(t), y(t)) \cdot y'(t)] dt$$

Green's theorem. D bounded, ∂D piecewise-smooth, $P, Q \in C^1$:

$$\int_{\partial D} P dx + Q dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy$$

Complex line integral. γ piecewise C^1 , $h(z)$ continuous:

$$\int_{\gamma} h(z) dz = \int_a^b h(x(t) + iy(t)) \cdot (x'(t) + iy'(t)) dt$$

Key integrals. $\gamma = \text{CCW unit circle}$:

$$\int_{\gamma} z^n dz = \begin{cases} 2\pi i & n = -1 \\ 0 & n \neq -1 \end{cases} \quad \int_{\gamma} \frac{dz}{z - z_0} = 2\pi i \cdot R \quad (R = \text{winding } \#)$$

ML-inequality.

$$\left| \int_{\gamma} h(z) dz \right| \leq M \cdot L, \quad M = \max_{z \in \gamma} |h(z)|, \quad L = \text{len}(\gamma)$$

Mean value property. u harmonic on $D \supseteq \overline{D(z_0, r)}$:

$$u(z_0) = \frac{1}{2\pi} \int_0^{2\pi} u(z_0 + r e^{i\theta}) d\theta$$

Holds for analytic functions as well.

Maximum principle. f nonconstant analytic $\implies |f(z)|$ has no
interior max; $\max_{\bar{D}} |f| = \max_{\partial D} |f|$. Analogously for nonconstant
harmonic u . *Pf.* Harmonic: if $u(z_0) = M = \max$, MVT forces
 $u \equiv M$ on circles around z_0 ; $\{u = M\}$ open, $\{u < M\}$ open by
continuity, D connected $\implies u \equiv M$. *Modulus:* let $h = e^{-i\theta} f$
($\theta = \arg f(z_0)$); $\Re h \leq |h| \leq M$, $\Re h(z_0) = M$; harmonic max
 $\implies \Re h \equiv M \implies \Im h = 0 \implies f$ const.

Complex Integration

FTC I. $f(z)$ continuous on D with analytic primitive $F(z)$,
 $F'(z) = f(z)$:

$$\int_{\gamma} f(z) dz = F(B) - F(A)$$

Path-independent; zero for closed curves. $f'(z) \equiv 0$ on $D \implies f$
constant.

FTC II. f analytic on simply connected $D \implies$ primitive
 $F(z) = \int_{\gamma} f(\zeta) d\zeta$ exists (γ any path from z_0 to z). Equivalently: f
has a primitive $\iff \oint_{\gamma} f dz = 0$ for every closed loop.

Cauchy's theorem. D bounded, f analytic on $D \cup \partial D$:

$$\int_{\partial D} f(z) dz = 0$$

Pf. $\oint f dz = \oint (u dx - v dy) + i \oint (v dx + u dy)$; Green's:
 $= \iint (-v_x - u_y) + i \iint (u_x - v_y) = 0$ by C-R.

Cauchy integral formula. f analytic on $D \cup \partial D$, $z \in D$:

$$f(z) = \frac{1}{2\pi i} \int_{\partial D} \frac{f(w)}{w - z} dw$$

$$f^{(m)}(z) = \frac{m!}{2\pi i} \int_{\partial D} \frac{f(w)}{(w - z)^{m+1}} dw, \quad m \geq 0$$

Cauchy's estimate. f analytic on $\overline{D(z_0, r)}$, $|f(z)| \leq M$ on
 $\partial D(z_0, r)$:

$$|f^{(m)}(z_0)| \leq \frac{m! M}{r^m}$$

Liouville's theorem. f entire and bounded $\implies f$ constant. *FTA*
proof: if degree- n poly has no root, $g = 1/f$ is bounded entire $\implies$
constant: contradiction.

Morera's theorem (converse of Cauchy's). f continuous on D ;
 $\int_{\partial R} f(z) dz = 0$ for every rectangle $R \subseteq D \implies f$ analytic on D .

Sequences and Series of Functions

Pointwise convergence. $(f_n(z))$ converges pointwise to $f(z)$ on E if $\lim_{n \rightarrow \infty} f_n(z) = f(z)$ for each $z \in E$.

Uniform convergence. $(f_n(z))$ converges uniformly to $f(z)$ on E , written $f_n(z) \Rightarrow f(z)$, if

$$\lim_{n \rightarrow \infty} \left(\sup_{z \in E} |f_n(z) - f(z)| \right) = 0$$

Normal convergence. $(f_n(z))$ converges normally to $f(z)$ on domain D if $f_n(z) \Rightarrow f(z)$ on every closed disk $\overline{D}(z_0, r) \subseteq D$.

Theorems.

- $(f_n(z))$ continuous, $f_n(z) \Rightarrow f(z)$ on $E \Rightarrow f(z)$ continuous
- $(f_n(z))$ analytic, $f_n(z) \Rightarrow f(z)$ on $D \Rightarrow f(z)$ analytic
- **Weierstrass:** $(g_n(z))$ analytic, $\sum g_n(z)$ converges normally to $g(z)$ on $D \Rightarrow g(z)$ analytic and $(\sum_{n=0}^{\infty} g_n(z))^{(k)} = \sum_{n=0}^{\infty} g_n^{(k)}(z)$ (term-by-term differentiation)

Weierstrass M-test. $|g_n(z)| \leq M_n$ on E ,

$\sum M_n < \infty \Rightarrow \sum g_n(z)$ converges uniformly on E .

Root test. $\sum a_k$ converges absolutely if $\limsup_k |a_k|^{1/k} < 1$; diverges if > 1 ; inconclusive if $= 1$.

Ratio test. $\sum a_k$ converges absolutely if $\lim_k |a_{k+1}/a_k| < 1$; diverges if > 1 ; inconclusive if $= 1$.

Power Series

Radius of convergence. For $\sum_{k=0}^{\infty} a_k(z - z_0)^k$:

$R = (\limsup_k \sqrt[k]{|a_k|})^{-1}$ (Cauchy-Hadamard);

$R = \lim_{k \rightarrow \infty} |a_k/a_{k+1}|$ (ratio test, if limit exists). Converges normally on $D(z_0, R)$; diverges outside $\overline{D}(z_0, R)$. Pf (C-H). Apply root test to $\sum a_k(z - z_0)^k$: $\limsup |a_k(z - z_0)^k|^{1/k} = |z - z_0| \cdot \limsup |a_k|^{1/k} < 1 \iff |z - z_0| < R$.

Analyticity. $f(z) = \sum_{k=0}^{\infty} a_k z^k$ is analytic on $|z| < R$, with $a_k = f^{(k)}(0)/k!$.

Taylor expansion. f analytic on $D(z_0, R)$:

$$f(z) = \sum_{k=0}^{\infty} \frac{f^{(k)}(z_0)}{k!} (z - z_0)^k, \quad a_k = \frac{1}{2\pi i} \oint_{|\zeta - z_0|=r} \frac{f(\zeta)}{(\zeta - z_0)^{k+1}} d\zeta$$

$|a_k| \leq M/r^k$ by Cauchy's estimate;

$R = \text{dist}(z_0, \text{nearest non-removable singularity})$ (poles/essential only; removable don't count since f extends analytically through them).

Power series arithmetic (both centered at z_0): Sum:

$\sum (a_k \pm b_k)(z - z_0)^k$. Product: $\sum c_k(z - z_0)^k$,

$c_k = a_0 b_k + a_1 b_{k-1} + \dots + a_k b_0$.

Schwarz lemma. f analytic on $\mathbb{D}$, $|f(z)| \leq 1$,

$f(0) = 0 \Rightarrow |f(z)| \leq |z|$ for all $z \in \mathbb{D}$. If $|f(z_0)| = |z_0|$ for some $z_0 \neq 0 \Rightarrow f(z) = e^{i\theta} z$.

Zeros. f analytic on D , $f(z_0) = 0$: either $f(z) \equiv 0$ on D , or z_0 is an isolated zero of order N (first nonzero a_N):

$f(z) = (z - z_0)^N g(z)$, $g(z_0) \neq 0$.

Identity theorem. $f(z), g(z)$ analytic on D ; if $f(z_k) = g(z_k)$ for a sequence z_k with $\lim_{k \rightarrow \infty} z_k = z_0 \in D \Rightarrow f(z) \equiv g(z)$ on D .

Laurent Series and Singularities

Laurent series. f analytic on $A_{\rho, \sigma}(z_0) = \{\rho < |z - z_0| < \sigma\}$ has a unique Laurent series converging normally to f on each such annulus: $f(z) = \sum_{k=-\infty}^{\infty} a_k(z - z_0)^k$

$$a_k = \frac{1}{2\pi i} \int_{\partial D(z_0, r)} \frac{f(w)}{(w - z_0)^{k+1}} dw, \quad r \in (\rho, \sigma)$$

Same f has different Laurent series on different annuli centered at z_0 .

Isolated singularity. f analytic on $\dot{D}(z_0, R) = A_{0, R}(z_0)$ but not at z_0 .

Removable singularity. $a_k = 0$ for all $k < 0$; equivalently: $\lim_{z \rightarrow z_0} f(z)$ exists; equivalently: f locally bounded. *Riemann's theorem:* $|f(z)| < M$ on $\dot{D}(z_0, \delta) \Rightarrow z_0$ removable.

Pole of order N . $a_{-N} \neq 0$, $a_k = 0$ for $k < -N$; equivalently:

$f(z) = g(z)/(z - z_0)^N$, g analytic, $g(z_0) \neq 0$; equivalently: $|f(z)| \rightarrow \infty$. For $f = P/Q$: pole order = mult. of zero of Q - mult. of zero of P . Principal part: $g(z) = (z - z_0)^m f(z)$ analytic $\Rightarrow a_{k-m} = g^{(k)}(z_0)/k!$.

Essential singularity. $a_k \neq 0$ for infinitely many $k < 0$.

Casorati-Weierstrass: $f(\dot{D}(z_0, r))$ is dense in $\mathbb{C}$, i.e. for any $w \in \mathbb{C}$ and $\varepsilon > 0$, $\exists z \in \dot{D}(z_0, r)$ s.t. $|f(z) - w| < \varepsilon$.

Residue Theory

Residue. $\text{Res}[f; z_0] = a_{-1} = \frac{1}{2\pi i} \oint_{|z - z_0|=r} f(z) dz$, $0 < r < R$

Residue theorem. f analytic on $D \setminus \{z_1, \dots, z_n\}$, extends continuously to ∂D :

$$\int_{\partial D} f(z) dz = 2\pi i \sum_{k=1}^n \text{Res}[f; z_k]$$

Pf. Excise disks D_k around z_k ; Cauchy on $D \setminus \bigcup \overline{D_k}$:

$\oint_{\partial D} = \sum_k \oint_{\partial D_k}$. Each $\oint_{\partial D_k}$: integrate Laurent termwise; only $n = -1$ survives ($\oint (z - z_k)^n dz = 2\pi i$ iff $n = -1$), giving $2\pi i a_{-1}$.

Four rules for computing residues.

Rule 1 (simple pole at z_0): $\text{Res}[f; z_0] = \lim_{z \rightarrow z_0} (z - z_0)f(z)$

Rule 2 (pole of order N at z_0):

$$\text{Res}[f; z_0] = \frac{1}{(N-1)!} \lim_{z \rightarrow z_0} \frac{d^{N-1}}{dz^{N-1}} [(z - z_0)^N f(z)]$$

Rule 3 ($f(z), g(z)$ analytic, $g(z)$ has simple zero at z_0):

$\text{Res}[f(z)/g(z); z_0] = f(z_0)/g'(z_0)$

Rule 4 ($g(z)$ analytic, simple zero at z_0): $\text{Res}[1/g(z); z_0] = 1/g'(z_0)$

Real integrals. P, Q polynomials, $\deg P \leq \deg Q - 2$, $Q(x) \neq 0$ on $\mathbb{R}$:

$$\int_{-\infty}^{\infty} \frac{P(x)}{Q(x)} dx = 2\pi i \sum_{\substack{Q(z_k)=0 \\ \Im z_k > 0}} \text{Res} \left[\frac{P(z)}{Q(z)}; z_k \right]$$

For $\cos(ax)/\sin(ax)$: replace with e^{iaz} , apply residue theorem, take $\Re/\Im$ part.

Jordan's lemma. $a > 0$, γ_R = upper semicircle: $|e^{iaz}| = e^{-a\Im z} \leq 1$ on γ_R (since $\Im z \geq 0$); $|\int_{\gamma_R} \frac{P}{Q} e^{iaz} dz| \rightarrow 0$ as $R \rightarrow \infty$ when $\deg Q > \deg P$.

Common Taylor Series

$$f(z) = \sum_{k=0}^{\infty} \frac{f^{(k)}(z_0)}{k!} (z - z_0)^k = f(z_0) + f'(z_0)(z - z_0) + \frac{f''(z_0)}{2!} (z - z_0)^2 + \dots$$

$$e^z = \sum_{k=0}^{\infty} \frac{z^k}{k!} = 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \dots$$

$$\sin z = \sum_{k=0}^{\infty} \frac{(-1)^k z^{2k+1}}{(2k+1)!} = z - \frac{z^3}{3!} + \frac{z^5}{5!} - \dots \quad \cos z = \sum_{k=0}^{\infty} \frac{(-1)^k z^{2k}}{(2k)!} = 1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \dots$$

$$\frac{1}{1-z} = \sum_{k=0}^{\infty} z^k = 1 + z + z^2 + z^3 + \dots \quad (|z| < 1)$$

$$\log(1+z) = \sum_{k=1}^{\infty} \frac{(-1)^{k-1} z^k}{k} = z - \frac{z^2}{2} + \frac{z^3}{3} - \dots \quad (|z| < 1)$$

$$\sinh z = \sum_{k=0}^{\infty} \frac{z^{2k+1}}{(2k+1)!} = z + \frac{z^3}{3!} + \frac{z^5}{5!} + \dots \quad \cosh z = \sum_{k=0}^{\infty} \frac{z^{2k}}{(2k)!} = 1 + \frac{z^2}{2!} + \frac{z^4}{4!} + \dots$$

Miscellaneous

Exact vs. closed. $Pdx + Qdy$ is closed if $P_y = Q_x$; exact if it has a primitive. Exact $\Rightarrow$ closed; on simply connected D : closed $\iff$ exact.

Fundamental theorem of algebra. Every nonconstant polynomial has a complex root; degree- n polynomial has exactly n roots (counting multiplicity).

Riemann mapping theorem. Any simply connected domain $D \neq \mathbb{C}$ is conformally equivalent to $\mathbb{D}$.

Winding number. $\int_{\gamma} \frac{dz}{z - z_0} = 2\pi i \cdot R$, R = number of CCW loops around z_0 .